

10 June 2020 (Teams)

① Find LHDE w e.c. of min. order...

a) $-3, -6, -12, -24, \dots$ $\frac{\dots}{\dots} = 2$

$$x_k = -3 \cdot 2^k$$

$$x = 2 \rightarrow e \cdot 2^k$$

char. eq. : $x - 2 = 0$

$$x_{k+1} - 2x_k = 0.$$

b) $0, 1, 1, 2, 3, 5, 8, 13, 21$ (Fibo).

$$\Rightarrow \text{not a } \frac{\dots}{\dots} = 2 \Rightarrow x_{k+2} = x_{k+1} + x_k$$

$$\Rightarrow x^2 - x - 1 = 0.$$

c) $4 \cdot \left(\frac{1}{5}\right)^k + 7 \cdot \left(\frac{1}{3}\right)^k$

$$x_1 = \frac{1}{5} \rightarrow e_1 \cdot \left(\frac{1}{5}\right)^k$$

$$x_2 = \frac{1}{3} \rightarrow e_2 \cdot \left(\frac{1}{3}\right)^k$$

$$\Rightarrow \left(x - \frac{1}{5}\right)\left(x - \frac{1}{3}\right) = 0 \Rightarrow x^2 - \frac{1}{3}x - \frac{1}{5}x + \frac{1}{15} = 0 \Rightarrow$$

$$\Rightarrow x^2 - \frac{8}{15}x + \frac{1}{15} = 0 \Rightarrow x_{k+2} - \frac{8}{15}x_{k+1} + \frac{1}{15} = 0$$

$$\Rightarrow 15x_{k+2} - 8x_{k+1} + 1 = 0.$$

② $\begin{cases} y'' - 2xy' + 4y = 0 \\ y(0) = 3 \\ y'(0) = 0 \end{cases} \rightarrow \text{sol} = \text{polym.}?$

$$y(x) = ax^2 + bx + c$$

$$y'(x) = 2ax + b$$

$$y''(x) = 2a$$

$$\Rightarrow 2a - 2x(2ax + b) + 4(ax^2 + bx + c) = 0$$

$$\Rightarrow 2a - 2bx + 4bx + 4c = 0$$

$$\Rightarrow 2a + 4c + 2bx = 0 \mid : 2$$

$$\Rightarrow a + 2c + bx = 0 \Rightarrow \boxed{b = 0}$$

$$\Rightarrow a + 2c = 0 \Rightarrow a = -2c$$

$$y(0) = 3 = c \Rightarrow \boxed{c = 3} \Rightarrow \boxed{a = -6}$$

$$\Rightarrow y(x) = -6x^2 + 3$$

check: $y' = -12x$

$$y'' = -12.$$

$$y'' - 2xy' + 4y = -12 -$$

$$-2 \cdot x \cdot (-12x) + 4(3 - 6x^2) = 0$$

$$y(0) = 3, y'(0) = 0$$

$$\Rightarrow y(x) = 3 - 6x^2.$$

③ $\dot{x} = -x + 2x^2 - x^3 = 0 \Rightarrow x(-1 + 2x - x^2) = 0$.

• phase portrait?

$x_1^* = 0$

• orbits?

$-x^2 + 2x - 1 = 0 \quad | \cdot (-1)$

• $\varphi(t, 0), \varphi(t, 3)$ properties?

$x^2 - 2x + 1 = 0$

• attractor? $\rightarrow$ basin.

$\Delta = 4 - 4 = 0$

$x_{2,3} = \frac{2}{2} = 1$

equil: $x_1^* = 0, x_2^* = x_3^* = 1$.

let $f(x) = -x(x^2 - 2x + 1) = -x(x-1)^2$

x	$-\infty$	0	1	∞
$f(x)$	$+$	0	$-$	$-$



orbits: $(-\infty, 0), \{0\}, (0, 1), \{1\}, (1, \infty)$

0 equil $\Rightarrow \varphi(t, 0) = 0$

$\varphi(t, 1) = 1$.

!

$\varphi(t, 3)$

$3 \in (1, \infty)$

$\rightarrow <$

$\Rightarrow \varphi(t, 3)$ strictly decreasing
 $\Rightarrow \lim_{t \rightarrow \infty} \varphi(t, 3) = 1$

$\Rightarrow (0)$ attractor.

$A_0 = \{ \eta \mid \lim_{t \rightarrow \infty} \varphi(t, \eta) = 0 \} = (-\infty, 0) \cup \{0\} \cup (0, 1) = (-\infty, 1)$

④ $\begin{cases} \dot{x} = x \\ \dot{y} = -4y \end{cases} \Rightarrow A = \begin{pmatrix} 1 & 0 \\ 0 & -4 \end{pmatrix}; f(x, y) = \begin{bmatrix} x \\ -4y \end{bmatrix}; Jf(x, y) = \begin{bmatrix} 1 & 0 \\ 0 & -4 \end{bmatrix}$

• type? stability sys. $\begin{vmatrix} 1-\lambda & 0 \\ 0 & -4-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(-4-\lambda) = 0 \Rightarrow$
 $\Rightarrow -4 - \lambda + 4\lambda + \lambda^2 = 0 \Rightarrow$

$\Rightarrow \lambda^2 + 3\lambda - 4 = 0 \Rightarrow \lambda + 4\lambda - \lambda - 4 = 0 \Rightarrow \lambda(\lambda + 4) - (\lambda + 4) = 0 \Rightarrow$

$\Rightarrow (\lambda + 4)(\lambda - 1) = 0 \Rightarrow \lambda \in \{1, -4\} \Rightarrow$ SADDLE, unstable.

$\frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} \Rightarrow \frac{-4y}{x} = \frac{dy}{dx} \Rightarrow -4y dx = x dy \Rightarrow -4 \frac{y}{x} dx = dy \Rightarrow$

$\Rightarrow -4 \int \frac{1}{x} dx = \int \frac{1}{y} dy \Rightarrow -4 \ln|x| = \ln|y| \Rightarrow$

$$\Rightarrow -4 \cdot \ln|x| = \ln|y| + C \Rightarrow -4 \ln x - \ln y = C.$$

$$\Rightarrow 4 \ln x + \ln y = C$$

$$\Rightarrow \ln x^4 \cdot y = C.$$

$$H(x, y) = -4 \ln x - \ln y$$

$$\boxed{\text{Def} \Rightarrow H(\varphi(t, \eta)) = H(\eta)}$$

We need the flow: Let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \Rightarrow \begin{matrix} x(0) = \eta_1 \\ y(0) = \eta_2 \end{matrix}$

$$\text{IVP: } \begin{cases} \dot{x} = x \Rightarrow \dot{x} - x = 0 \\ \dot{y} = -4y \Rightarrow \dot{y} + 4y = 0 \\ x(0) = \eta_1 \\ y(0) = \eta_2 \end{cases}$$

$$\begin{matrix} \dot{x} - x = 0 & x(0) = c = \eta_1 \\ x - 1 = 0 \Rightarrow x = 1 \Rightarrow x(t) = c \cdot e^t, c \in \mathbb{R} \end{matrix} \} \Rightarrow x(t) = \eta_1 \cdot e^t$$

$$\begin{matrix} \dot{y} + 4y = 0 & y(0) = c = \eta_2 \\ x + 4 = 0 \Rightarrow x = -4 \Rightarrow y(t) = c \cdot e^{-4t} \end{matrix} \} \Rightarrow y(t) = \eta_2 \cdot e^{-4t}.$$

$$\varphi(t, \eta) = \varphi(t, \eta_1, \eta_2) = (\eta_1 \cdot e^t, \eta_2 \cdot e^{-4t})$$

$$H(\varphi(t, \eta)) = \ln((\eta_1 \cdot e^t)^4 \cdot \eta_2 \cdot e^{-4t}) = \ln(\eta_1^4 \cdot \eta_2) = H(\eta).$$

⑤ a) Represent in the complex plane the curve:

$$\{ 2 \cdot i \cdot e^{-2t+4it} : t \geq 0 \}$$

$$\Rightarrow \begin{matrix} r(t) = 2 \cdot e^{-2t} \\ \theta(t) = 4t - \frac{\pi}{2} \end{matrix}$$

$$z(t) = -2 \cdot i \cdot e^{-2t} \cdot e^{4it} = 2i \cdot e^{-2t} \cdot e^{i \cdot 4t} = r \cdot e^{i\theta}$$

$$= -2i \cdot e^{-2t} \cdot (\cos 4t + i \sin 4t) = \underbrace{2 \cdot e^{-2t} \cdot \sin 4t}_{x(t)} - \underbrace{2i \cdot e^{-2t} \cdot \cos 4t}_{y(t)}$$

$$\begin{cases} x(t) = 2e^{-2t} \cdot \sin 4t \\ y(t) = -2e^{-2t} \cdot \cos 4t \end{cases} = 2e^{-2t} (\sin 4t - i \cos 4t)$$

$$t=0 \Rightarrow z(0) = -2i \Rightarrow (0, -2i) = (t, z(t))$$

$$\arg(z(t)) = |z(t)| = (2 \cdot e^{-2t}) \Rightarrow \lim_{t \rightarrow \infty} |\arg(z(t))| = \lim_{t \rightarrow \infty} 2 \cdot \frac{1}{e^{2t}} = 0. \Rightarrow \text{spiral decreases.} \quad (\text{radius})$$

$$\arg(z(t)) = 4t - \frac{\pi}{2} \Rightarrow \text{spiral through origin} \Rightarrow \text{counterclockwise}$$

$$\arg'(z(t)) = 4 > 0$$

$\Rightarrow$ angle increases.



$$b) \begin{cases} \dot{x} = -2x - 4y \\ \dot{y} = 4x - 2y \\ x(0) = 0 \\ y(0) = -2 \end{cases}$$

orbit? phase plane.

LINEAR $\Rightarrow$ NO JACOBIAN.

$(0,0)$ equil. (unique)

$$A = \begin{pmatrix} -2 & -4 \\ 4 & -2 \end{pmatrix} \Rightarrow \begin{vmatrix} -2-\lambda & -4 \\ 4 & -2-\lambda \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow (-2-\lambda)^2 + 16 = 0 \Rightarrow (2+\lambda)^2 + 16 = 0 \Rightarrow 4 + 4\lambda + \lambda^2 + 16 = 0$$

$$\Rightarrow \lambda^2 + 4\lambda + 20 = 0 \Rightarrow \lambda_{1,2} = -2 \pm 4i \Rightarrow \text{Focus} \Rightarrow \text{stable. asymp.}$$

$$a) \Rightarrow \gamma(0, -2) = \{ (2 \cdot e^{-2t} \sin 4t, -2e^{-2t} \cos 4t) : t \in \mathbb{R} \}.$$

$$(1,0) \Rightarrow \begin{cases} \dot{x} = -2 \\ \dot{y} = 4 \end{cases} \Rightarrow \begin{matrix} < \\ > \end{matrix} \Rightarrow \text{counterclockwise spirals, inward to } (0,0) \text{ equil. attr.}$$

c); b) $\Rightarrow$ stable. focus.

$(0,0)$ global attractor $\Rightarrow \nexists H$.

$$\frac{\dot{y}}{\dot{x}}$$

$$d) \lim_{t \rightarrow \infty} e^{At}, \quad A = \begin{pmatrix} -2 & -4 \\ 4 & -2 \end{pmatrix} = 2 \cdot \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$$

$$\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$$

$$\operatorname{Re}(\lambda) < 0 \Rightarrow e^{At} \rightarrow 0$$

$$\rightarrow \lim_{t \rightarrow \infty} e^{At} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A' = A$$

$$A^2 = \begin{pmatrix} -2 & -4 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} -2 & -4 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} -12 & -16 \\ -16 & -12 \end{pmatrix}$$

~~A^3~~